Electronic and Telecommunication Engineering

Section 1

Introduction to Medical Imaging

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Acknowledgement: Dr. Nuwan Dayananda for the content

Overview of Medical Imaging

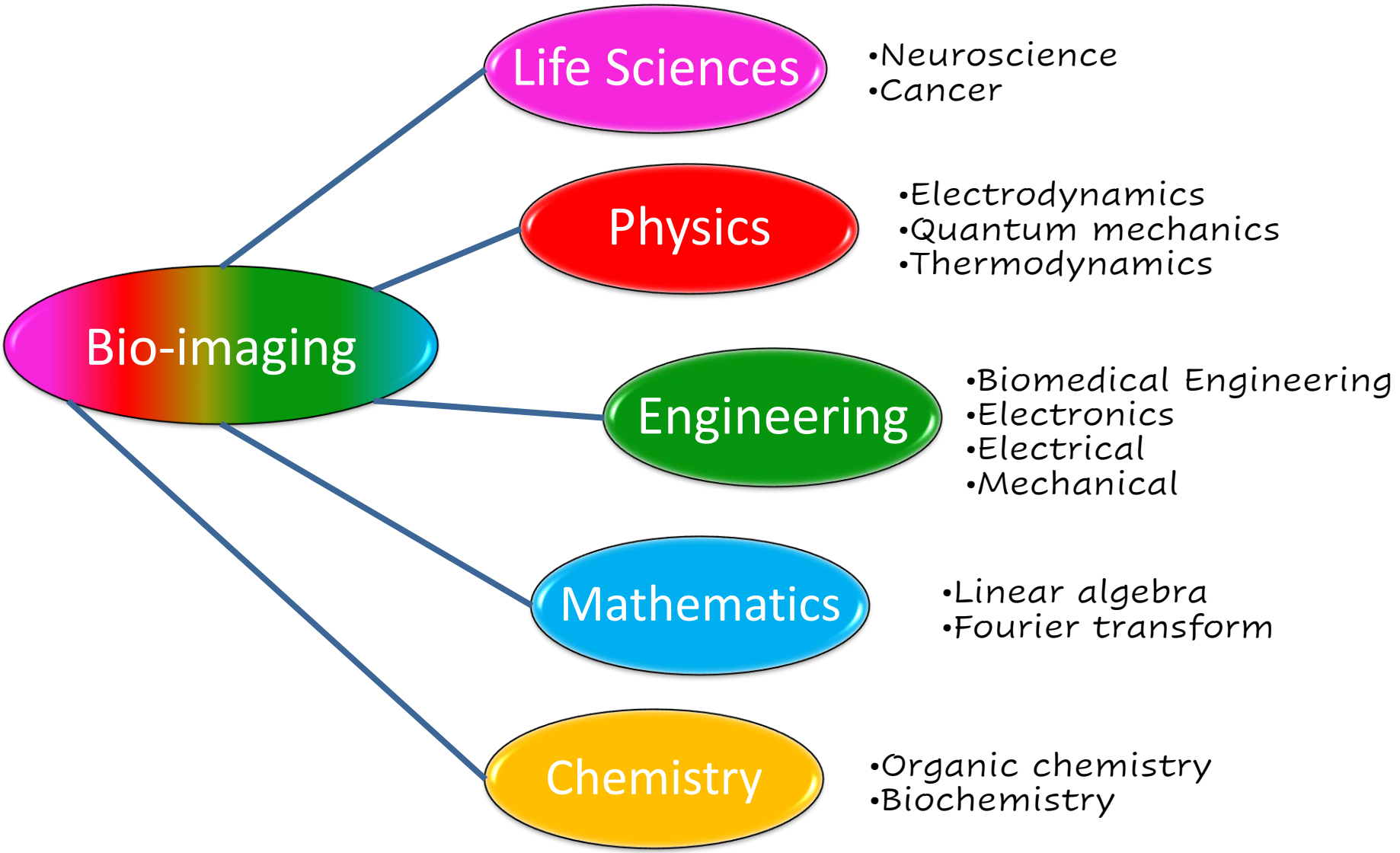
What is Medical Imaging?

Several different technologies that are used to **view the human body** in order to **diagnose, monitor, or treat** medical conditions.

Images of Human Body

- Derived from the interaction of energy with human tissue
  - Examples: radiation, magnetic, electric, or acoustic energy
  - Interactions usually at the molecular or atomic levels

Bio-imaging is built upon..



Imaging Modalities

- Different techniques for acquiring an image of the body
  - Projection X-ray (Radiography)
  - CT: Computed (or Computerized) Tomography
  - MR, or MRI: Magnetic Resonance Imaging
  - Ultrasound: Sonar in the body
  - PET: Positron Emission Tomography
  - SPECT: Single Photon Emission Tomography
  - OCT: Optical Coherence Tomography (infrared)

Myth Busting

- How much do you already know about medical imaging?

True or False?

- You are exposed to X-ray whenever you enter a room with an X-ray machine.

False

True or False?

- X-rays do not stay in the body after the radiation exposure.

True

True or False?

- Pregnant women should never have an X-ray.

False

True or False?

- CT is like a computerized X-ray.

True

True or False?

- CAT stands for Computer-Aided Tomography.

True

True or False?

- CT is only used to diagnose cancer.

False

True or False?

- Ultrasound machines use mechanical waves to produce an image.

True

True or False?

- Ultrasound images are always 2D.

False

True or False?

- Ultrasound can be used to locate tumours.

True

True or False?

- Magnetic Resonance Imaging (MRI) uses ionizing radiation to produce an image.

False

True or False?

- MRI is the safest imaging diagnostic tool for everyone.

False

True or False?

- It is a good idea to leave all jewellery at home when you go for an MRI.

True

True or False?

- A 3T MRI scanner, commonly used in clinical practice, has 60,000 times the force of the Earth's magnetic field.

True

True or False?

- If you get injected with a radio-pharmaceutical for a PET scan, you become radioactive.

True (only temporary)

True or False?

- Nuclear medicine that uses PET and SPECT scanners is only used for oncology.

False

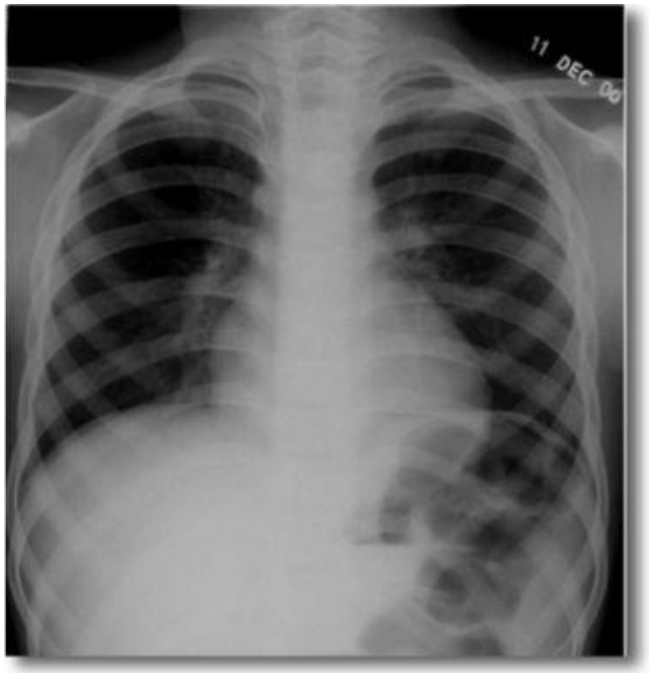
Imaging Modalities

- Why we need different imaging modalities?
  - Different types of **information**
  - Different **acquisition** procedures
  - Different **effects on the body**
  - **Complementary** information

X-ray radiography

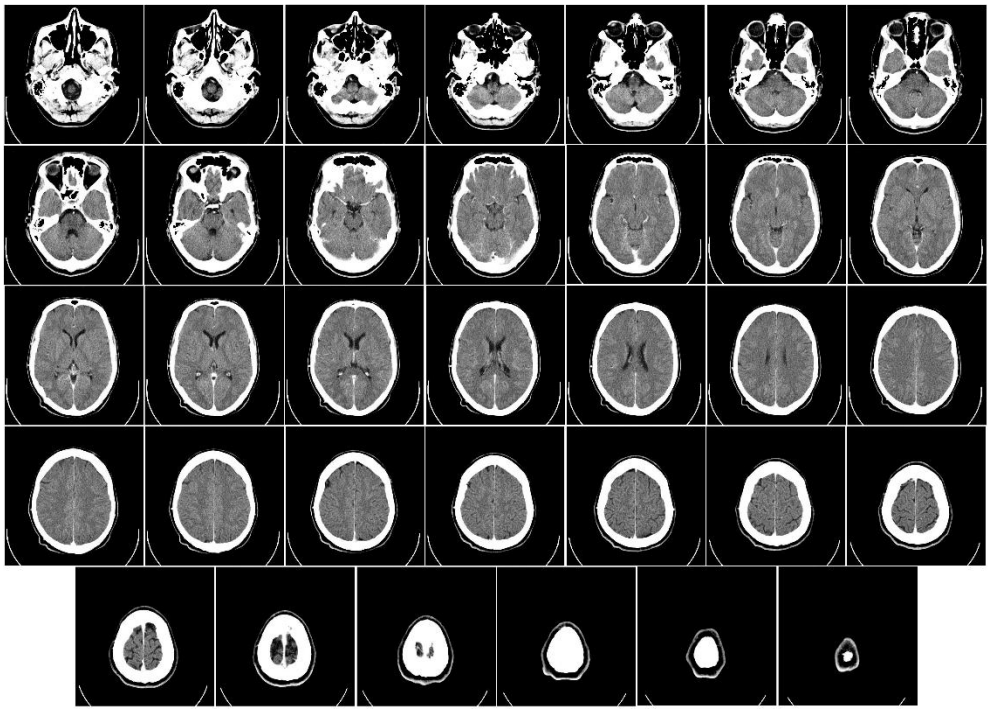
- **Electromagnetic radiation**
- High energy photons **penetrate** the body, but are **absorbed** by some tissues
- A **projection** technique

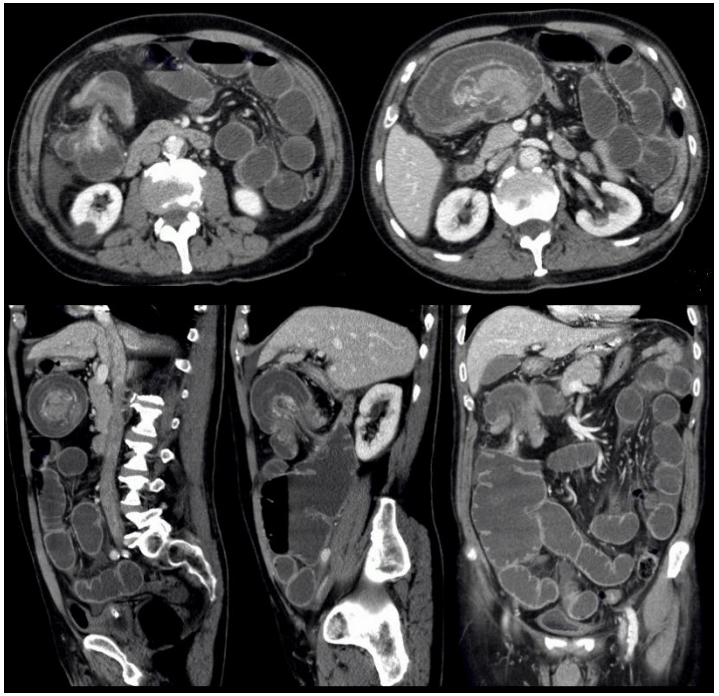




CT Imaging

- Uses **X-rays**
- Can selects a **thin slice of tissue**
- Needs **lots of projections** from different directions
- Can build a **3D volume**





MR Imaging

- Uses **magnetization** of Hydrogen atoms in the body
- Uses **RF electromagnetic wave to excite** the Hydrogen atoms
- Can produce many different types of **contrast**

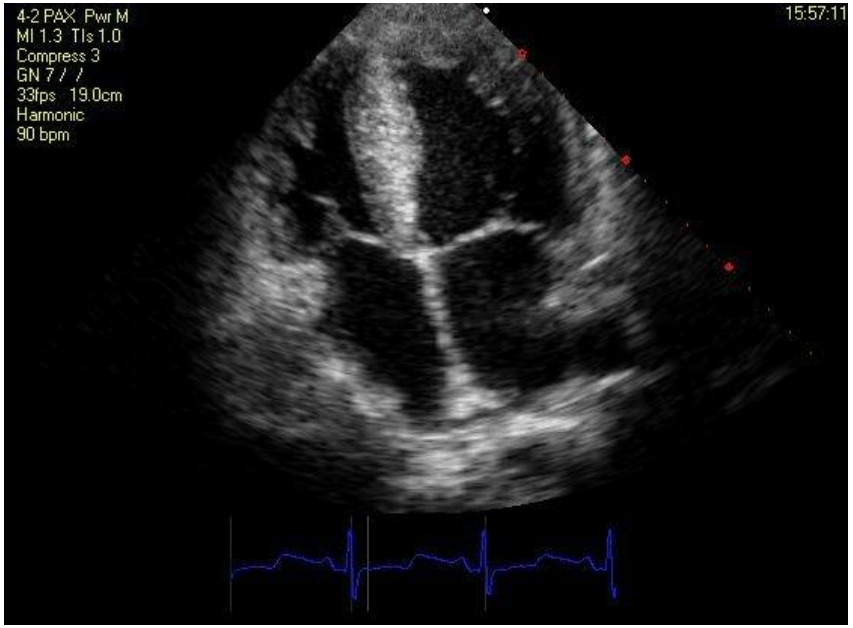




Ultrasound Imaging

- Uses **sound waves**, that are **reflected** by the body tissues
- Uses a **transducer to transmit sound waves** and then to collect the signal that is **reflected**

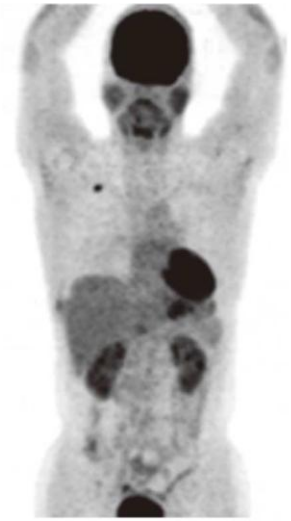


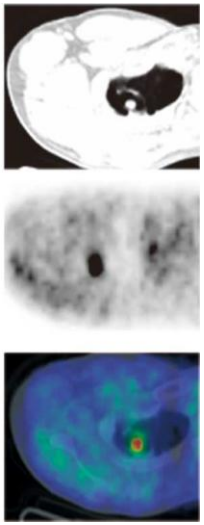


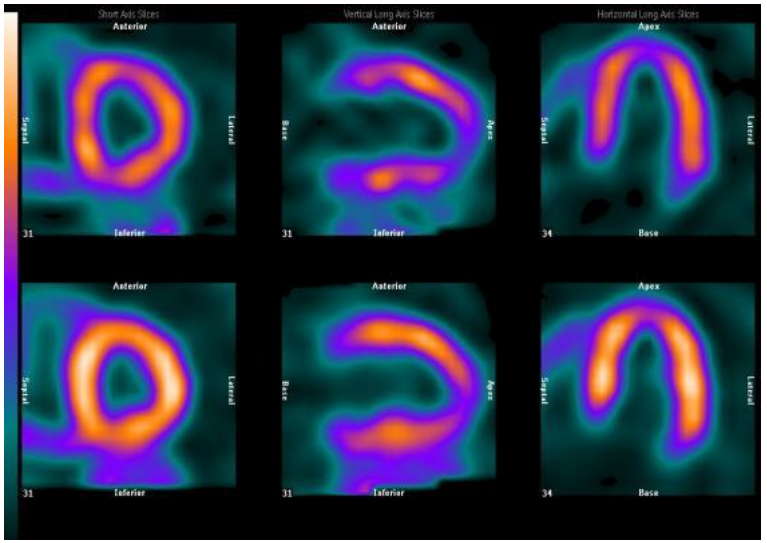
Nuclear Imaging

- Two types:
  - SPECT: Single Photon Emission Tomography
  - PET: Positron Emission Tomography
- Uses small amount of **radioactive tracers**
- Detects **gamma rays** emitted during the decay









Categorization of Imaging Modalities

- Modalities that cause ionizing radiation
  - X-ray
  - CT
  - PET
  - SPECT
- Non-ionizing modalities
  - MRI
  - Ultrasound
  - OCT

Categorization of Imaging Modalities cont.

- Structural

Tissue density, region size, shape, and orientation

  - X-ray
  - CT
  - MRI
  - Ultrasound

- Functional

Activity (metabolic rate), perfusion, ventilation

  - PET
  - SPECT
  - Functional MRI (fMRI)
  - Doppler Ultrasound

- Challenge
  - Combining structural and functional information together

What do images represent?

- For medical images, pixels can represent parameters such as:
  - X-ray attenuation (density) → X-ray, CT
  - Water (proton) density → MRI
  - Acoustic impedance → Ultrasound
  - Optical reflectivity (or impedance) → OCT
  - Tracer uptake (radioactivity) → PET, SPECT
  - Heat → Thermal imaging

Image Quality Measures

Image Quality Measures

- Directly related to the parameters which are important to a clinician for diagnosis

- Most important quantitative measures of image quality
  - Spatial resolution
  - Signal-to-Noise Ratio (SNR)
  - Contrast-to-Noise Ratio (CNR)

Image Quality Measures cont.

- Example: a good quality image
  - An MR image showing two small white-matter lesions
  - Image quality is high enough to detect and resolve the two lesions.

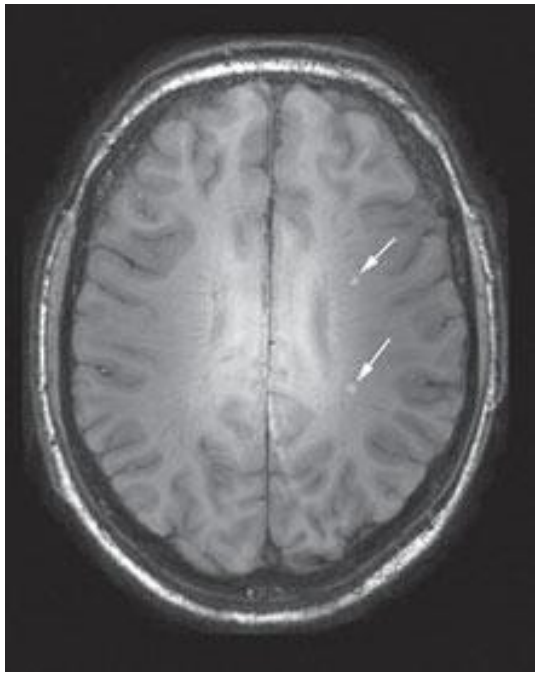
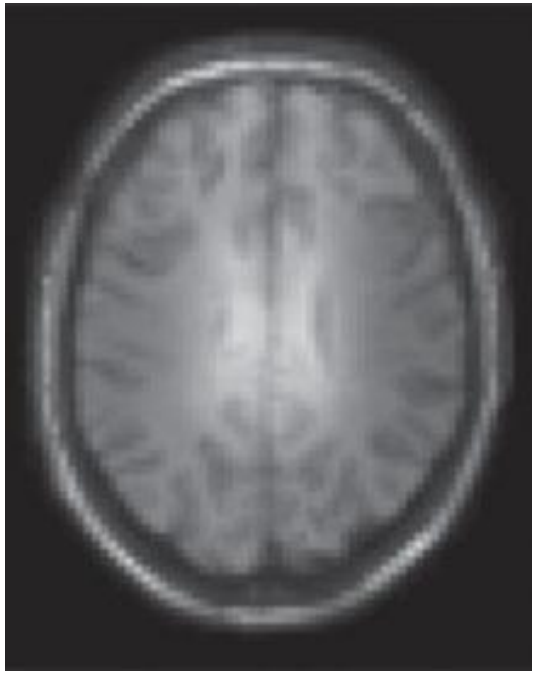
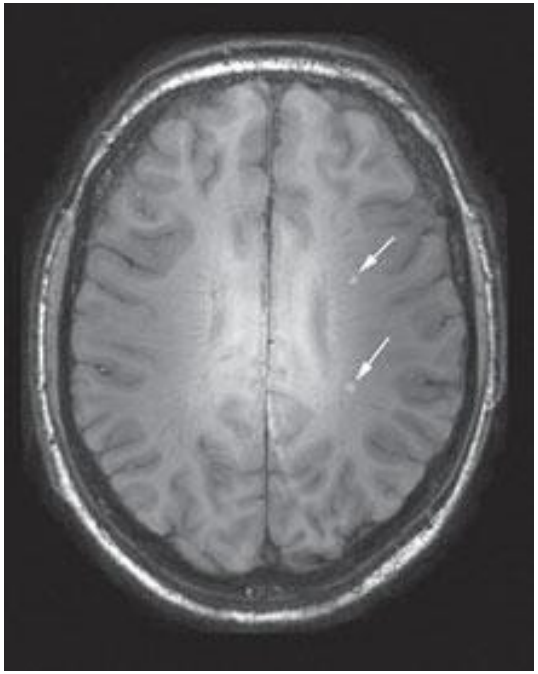


Image Quality Measures cont.

- Example: four times reduced resolution image
  - Only the larger of the two lesions is barely visible



Reduced resolution



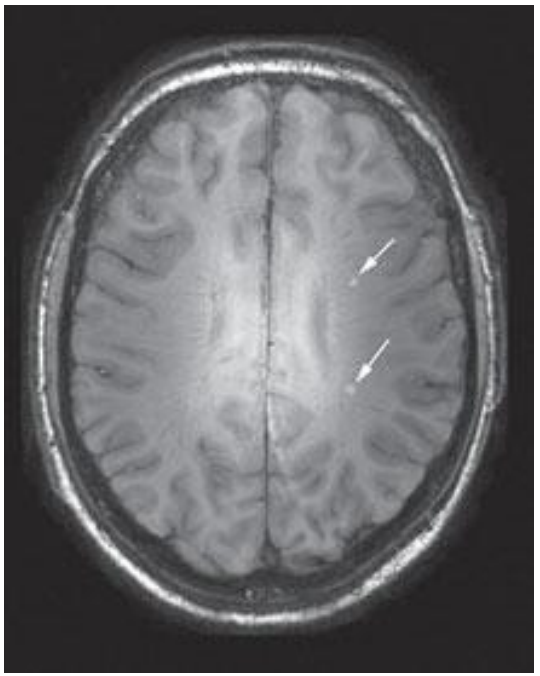
Original

Image Quality Measures cont.

- Example: four times reduced SNR image
  - Only the brighter of the two lesions is barely visible



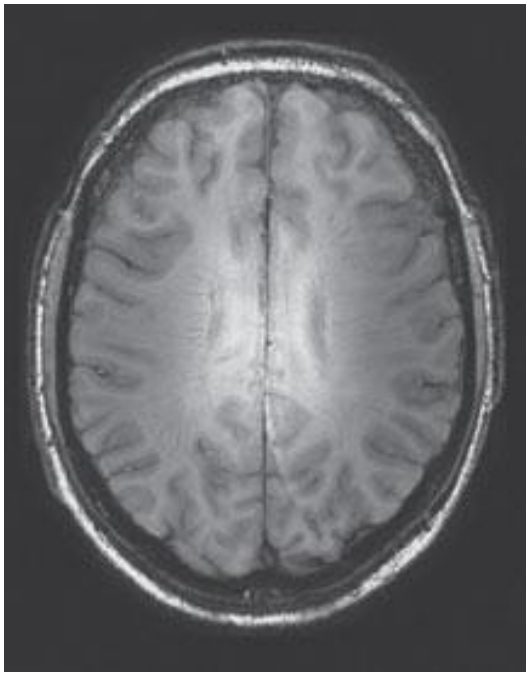
Reduced SNR



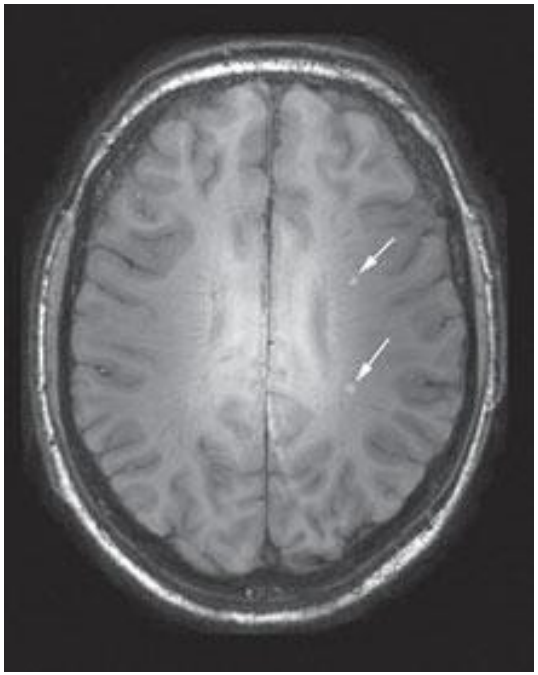
Original

Image Quality Measures cont.

- Example: reduced CNR between the lesions and the surrounding healthy tissue
  - Both lesions are not detectable



Reduced CNR



Original

Spatial Resolution

Spatial Resolution

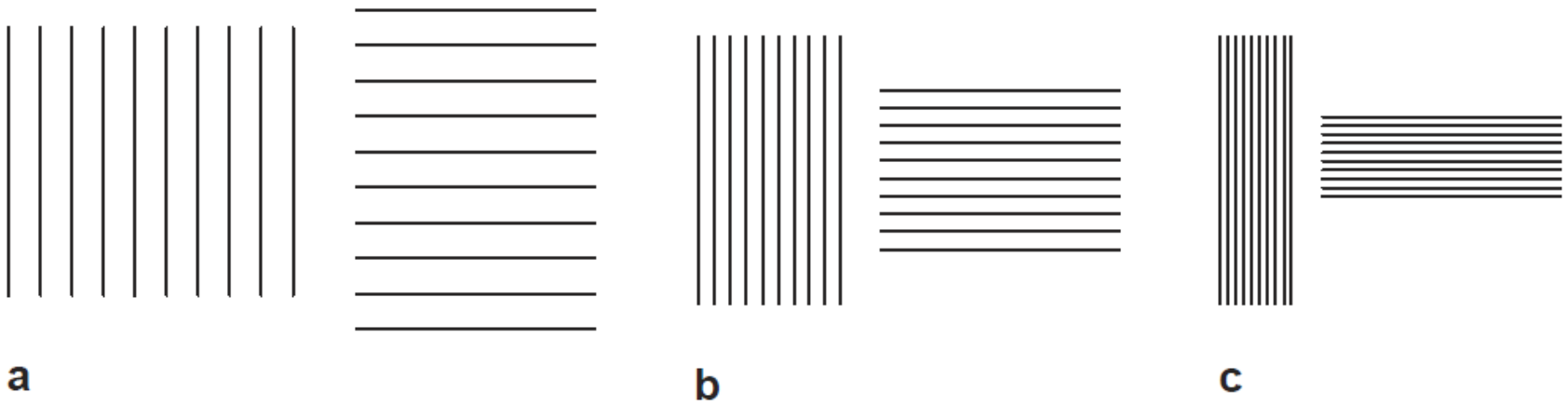
- **Smallest distance** between two features such that the features can be individually resolved

- Common measures in the spatial domain
  - Line Spread Function (LSF)
  - Point Spread Function (PSF)

- Equivalent representation in spatial frequency domain
  - Modulation Transfer Function (MTF)

Spatial Frequency

- Example: an optician`s test of a series of black lines on a white background
  - Lines are of different thickness and separation
  - Spatial frequency of a particular grid of lines is measured as the number of **lines/mm**
  - Example: **5 mm^{-1}** spatial frequency for lines spaced **200 µm** apart.



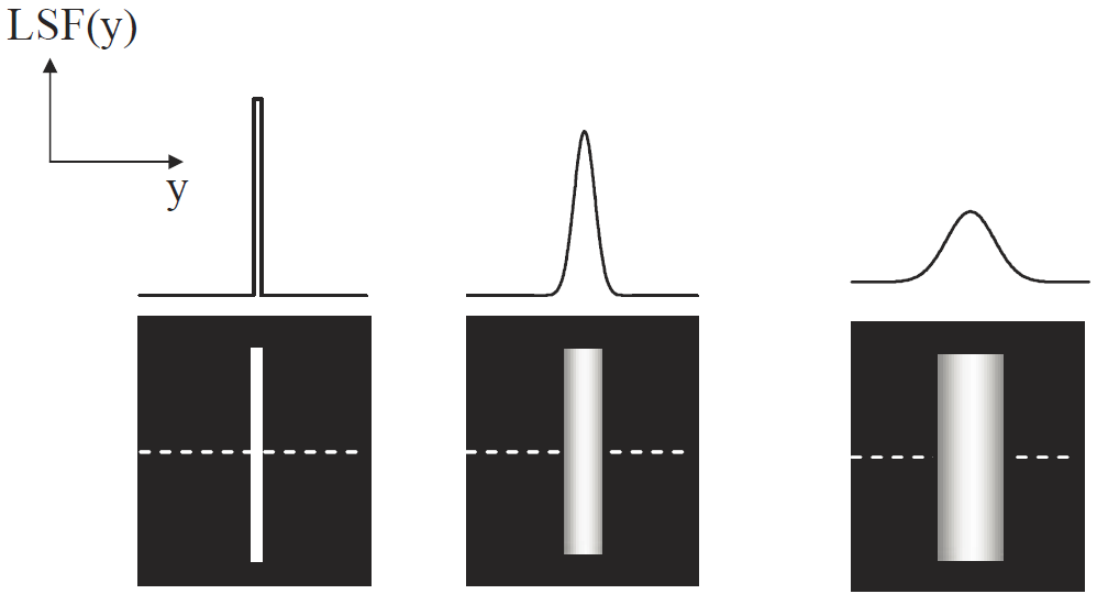
Grid patterns with increasing spatial frequencies

The Line Spread Function (LSF)

- Represents the degree of blurring in the image
- Describes system performance in one dimension
- Measures the spatial resolution of an imaging system:
  - A thin line or set of lines made of the appropriate material is imaged
    - Examples: a strip of lead for an X-ray scan or a thin wire embedded in gel for an ultrasound scan
  - Imaging system introduces some degree of blurring onto the line(s) in the image

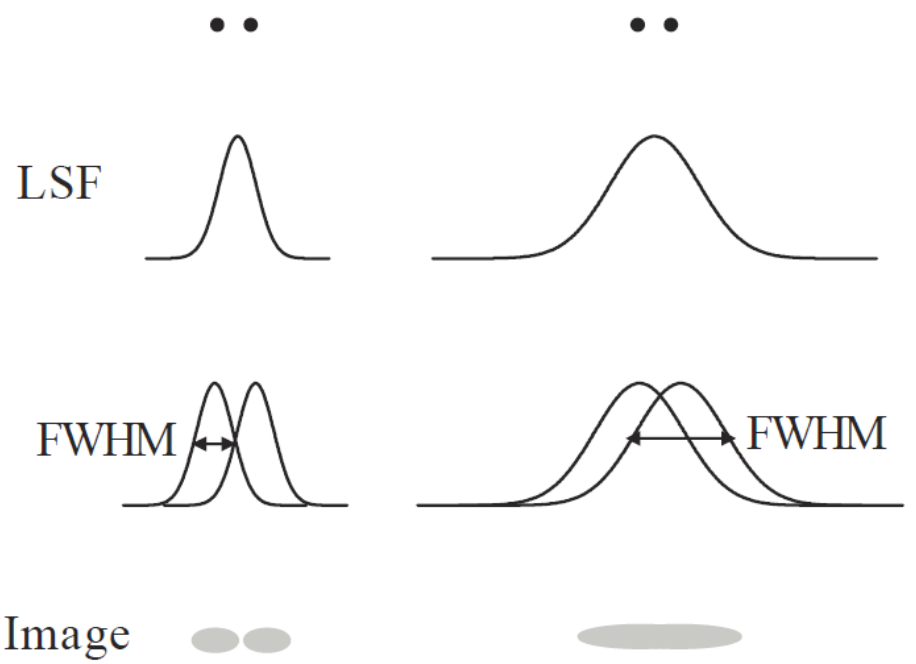
The Line Spread Function (LSF) cont.

- Example: A thin object is imaged using three different imaging systems of the same modality
  - One-dimensional projection measured along the dotted line as shown defines the sharpness of the LSF
  - The system on the left has the sharpest LSF



The Line Spread Function (LSF) cont.

- Estimated by measuring a one-dimensional projection (dotted line in the previous image)
  - Defined by a parameter known as the **Full-Width-at-Half Maximum (FWHM)**
  - If two small structures are separated by a distance greater than the FWHM of the LSF, then they can be resolved as separate structures
  - Example: image produced by two different systems with a narrow (left) and broad (right) LSF



The Line Spread Function (LSF) cont.

- For many imaging techniques, LSF is well-approximated by a Gaussian function;

$$LSF\ (x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-(x - x_0)^2}{2\sigma^2}\right)$$

where σ = standard deviation and x_0 = centre of the function

- FWHM of a Gaussian function is given by:

$$FWHM = \left(2\sqrt{2\ln 2}\right)\sigma \approx 2.36\,\sigma$$

- Two structures could be resolved if,

  physical separation > 2.36 σ of the LSF distribution

The Point Spread Function (PSF)

- A full description of the spatial resolution in all directions
- Describes the image acquired from a very small '**point source**'
- Three-dimensional (3D) image (I) and object (O) are related by:

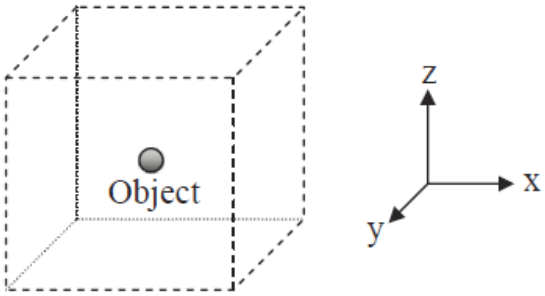
$$I\,(x,y,z) = O\,(x,y,z) * h\,(x,y,z)$$

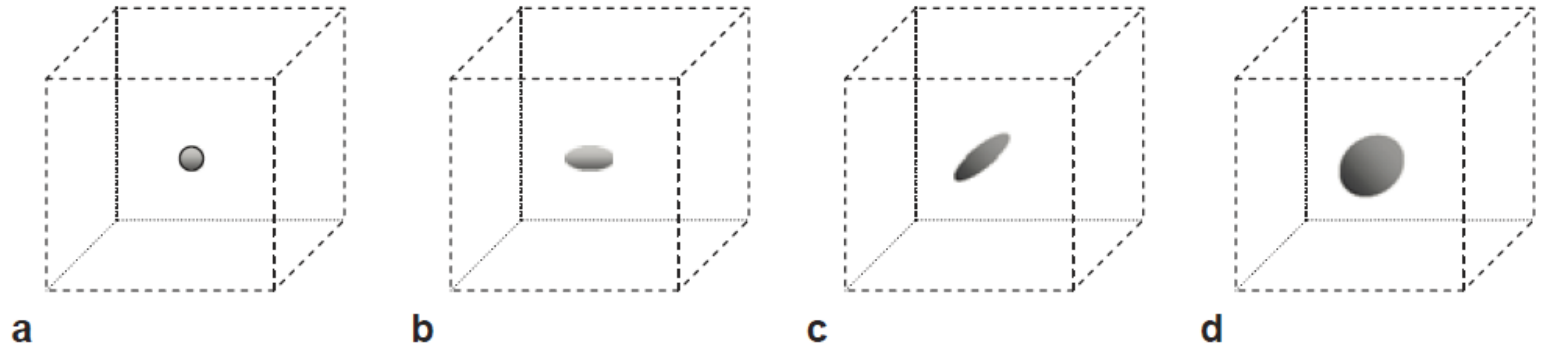
where $h\,(x,y,z)$ = 3D PSF

- Ideal system: PSF is a **delta function**
- In practice: overall PSF is a combination of **detection and data sampling circuits**

The Point Spread Function (PSF) cont.

- Examples of spatially symmetric and asymmetric PSFs:
  A small point object being imaged on the system having;
  - a: a **sharp** PSF in all three directions
  - b: a PSF significantly **broader in x** than in y or z
  - c: a PSF **broadest in the y**-direction
  - d: a PSF **broad in all three** directions





The Modulation Transfer Function (MTF)

- The most commonly used measure of the spatial resolution
- **Frequency response** of the imaging system
  - Ideal system should have an **infinitely uniform** MTF
  - But in practice, high spatial frequencies are **attenuated**.

    Greater attenuation → **Poorer spatial resolution**

- MTF is given by the Fourier transform of the PSF

$$\mathrm{MTF}\left(k_x, k_y, k_z\right) = F\left\{\mathrm{PSF}\left(x, y, z\right)\right\}$$

where k_x , k_y and k_z are the **spatial frequencies** measured in **lines/mm** corresponding to the *x, y* and *z* **spatial dimensions** measured in **mm**.

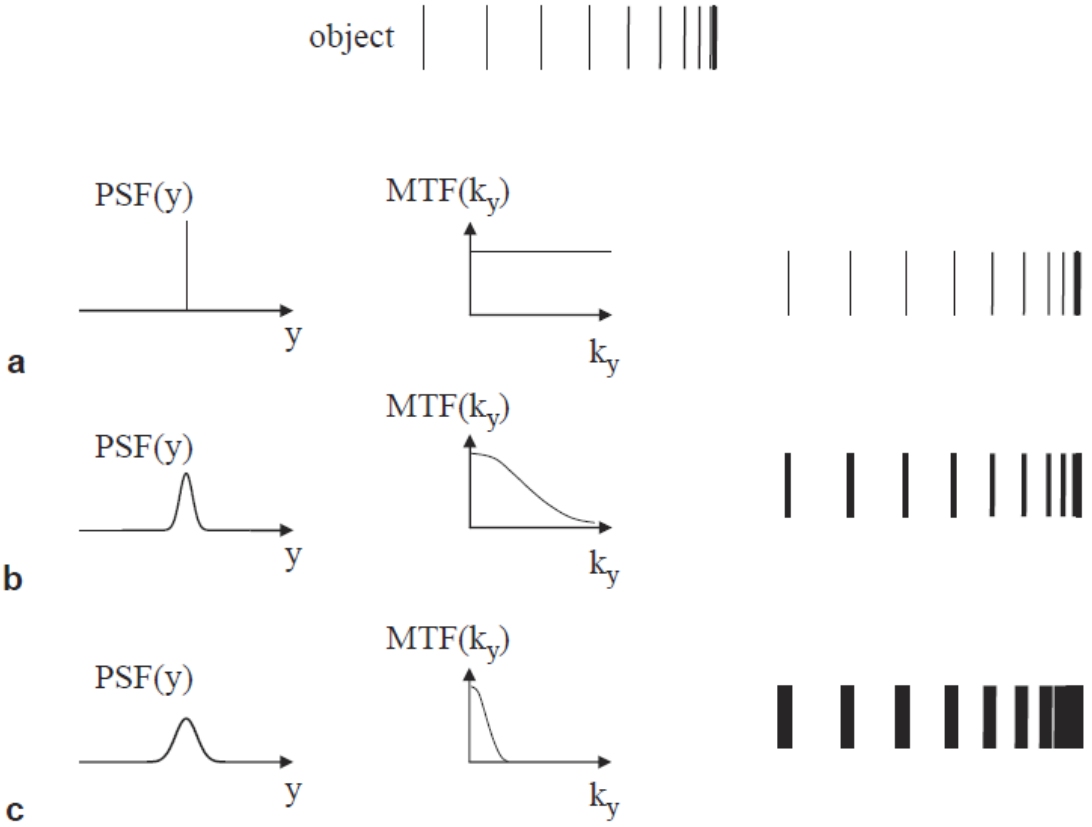
The Modulation Transfer Function (MTF) cont.

- Example: relationship between a one-dimensional MTF and PSF;

a. An **ideal** PSF and the corresponding MTF produce a **perfect** image

b. A **slightly broader** PSF produce an MTF which loses the **very high** spatial frequency information

c. The **broadest** PSF corresponds to the narrowest MTF, and the greatest loss of **high** spatial frequency information



Signal-to-noise ratio (SNR)

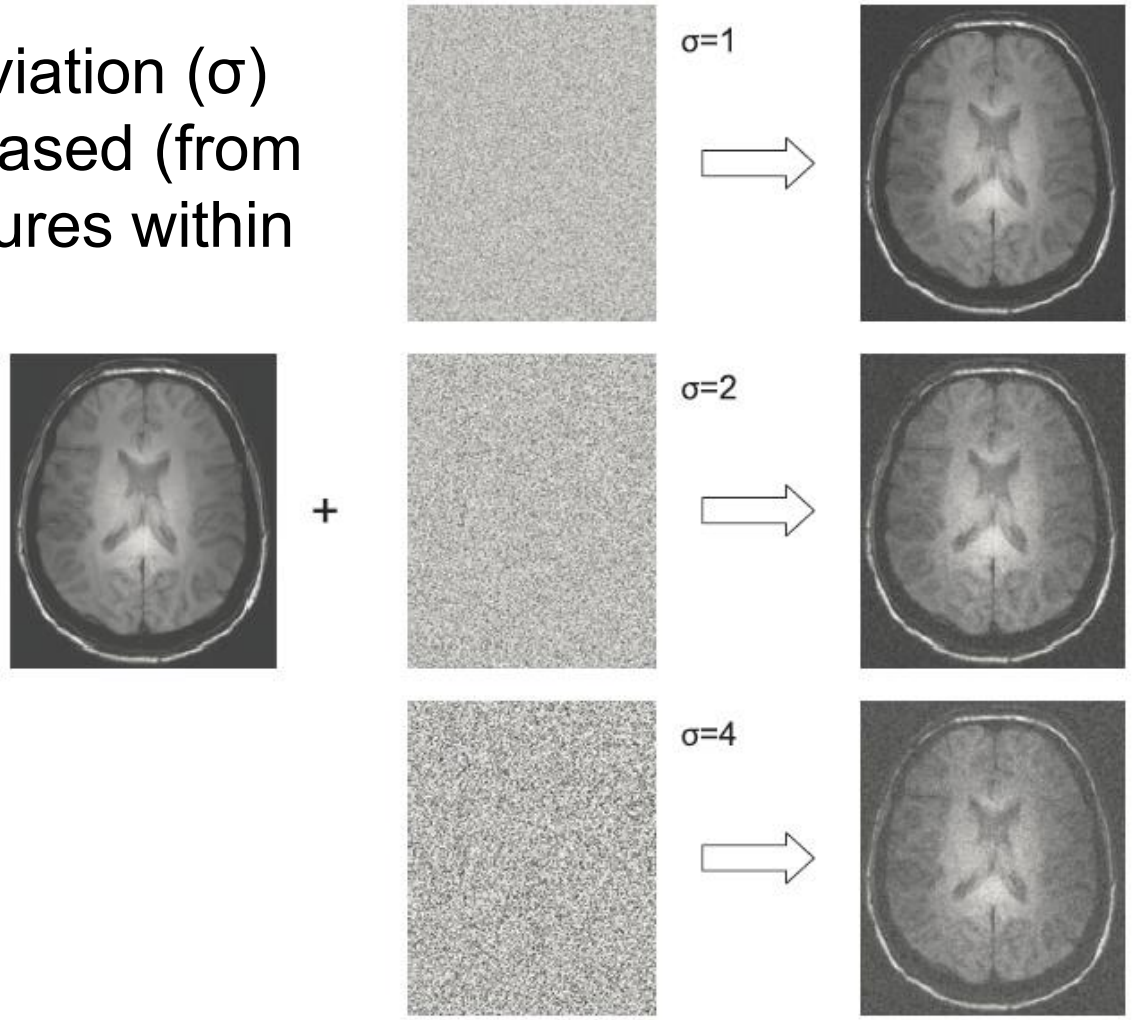
Signal-to-noise ratio (SNR)

- In all measured or recorded signals there is some contribution from noise.
  - Any signal that is **not related** to the actual signal being recorded
  - Does not include image **artifacts**
  - Often a **random** signal superimposed on top of the real signal
  - **Mean value** is zero which gives no indication of the noise level
  - Conventionally quantitative measure of the noise level is the **standard deviation**

- In medical imaging instrumentation
  - Recorded signal should be as **large** as possible in order to get the highest signal-to-noise ratio (SNR)
  - As the noise level increases, the information content and diagnostic utility of the image are reduced significantly.

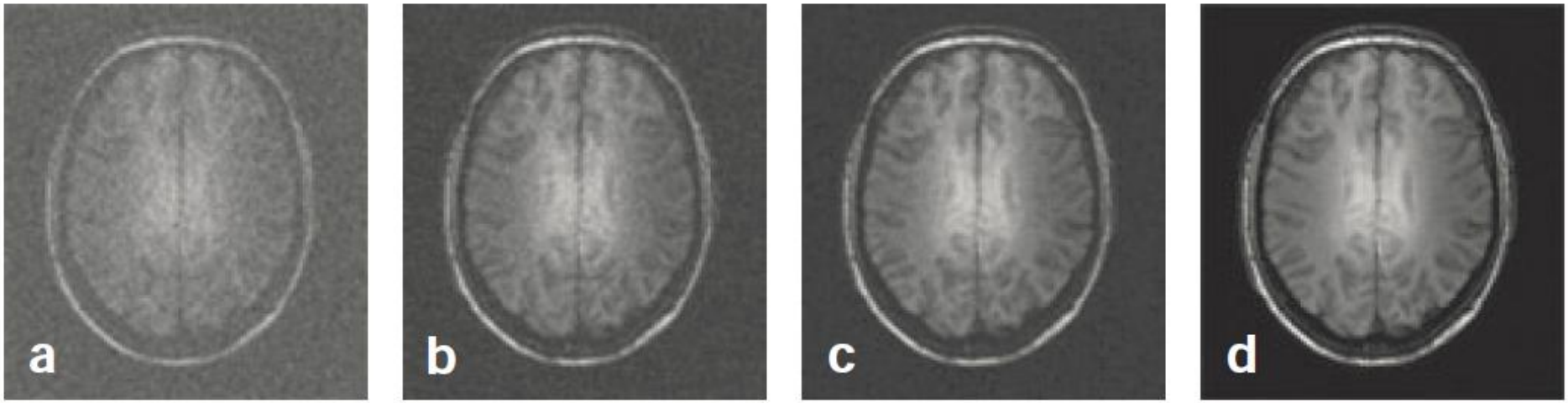
Signal-to-noise ratio (SNR) cont.

- Example: effects of noise on image quality for an MR image
  - As the standard deviation (σ) of the noise is increased (from top-to-bottom), features within the image become indistinguishable.



Signal-to-noise ratio (SNR) cont.

- For truly random noise as in MRI
  - Image SNR can be improved by **averaging** a number of scans together.
  - Example: Signal averaging to improve the image SNR. (a) MR image acquired in a single scan, (b) two scans, (c) four scans, and (d) sixteen scans averaged together.
  - Trade-off is the additional time required for data acquisition, hence, cannot be used in **dynamic** imaging.



Signal-to-noise ratio (SNR) cont.

- In ultrasound imaging
  - Signal averaging does not increase the SNR since major noise contribution from **speckle** is **coherent**
  - If images are acquired with the transducer oriented at different angles (compound imaging), averaging can sightly increase the SNR

- In X-ray and nuclear imaging
  - SNR is proportional to the square root of **photons detected**
  - The ultimate limit to the SNR is the radiation **dose to the patient**
  - X-ray images in general have very high SNR, but for nuclear medicine scans the number of γ-rays detected is much lower.

Contrast-to-noise ratio (CNR)

Contrast-to-noise ratio (CNR)

- Should be high enough to distinguish between different tissues

- A common definition of **contrast**:

$$C_{AB} = |S_A - S_B|$$

where C_{AB} is the contrast between tissues A and B, and S_A and S_B are the corresponding signals

- The CNR between tissues A and B is defined by:

$$CNR_{AB} = \frac{C_{AB}}{\sigma_N} = \frac{|S_A - S_B|}{\sigma_N} = |SNR_A - SNR_B|$$

where σ_N is the standard deviation of the noise.

Contrast-to-noise ratio (CNR) cont.

- CNR depends on:
  - Intrinsic **contrast** between particular tissues
  - Image **SNR**
  - Spatial resolution

- CNR can be reduced due to '**partial volume**' effects
  - Example: contrast between the lesion and healthy tissue is decreased, since voxels (volumetric pixels) contain contributions from both tissues

Image Artifacts

Image Artifacts

- Any signal in an image;
  - which is caused by a phenomenon related to the imaging process
  - But distorts the image or introduces an apparent feature which has no physical counterpart

- Recognizing the causes of artifacts is very important images interpretation

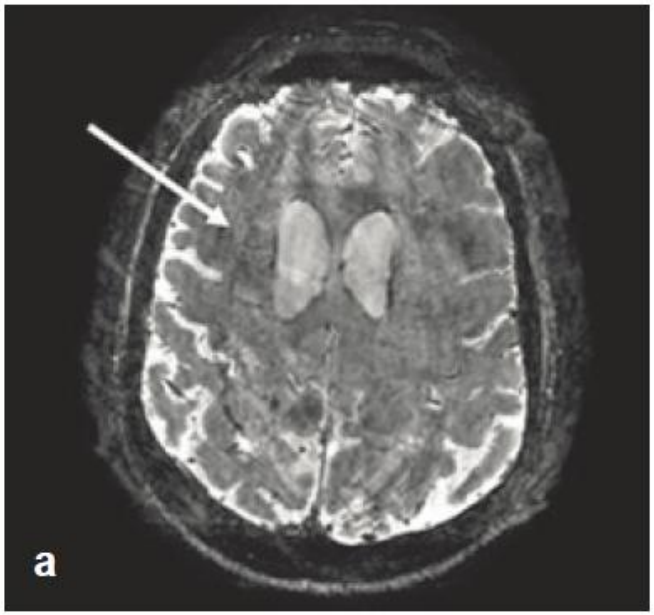
Image Artifacts cont.

- Examples of image artifacts

a: Motion in MRI causes extra lines to appear in the image

b: Acoustic shadowing in ultrasound produces a black hole in the image

c: A metal implant causes `streaking artifacts` in a CT image



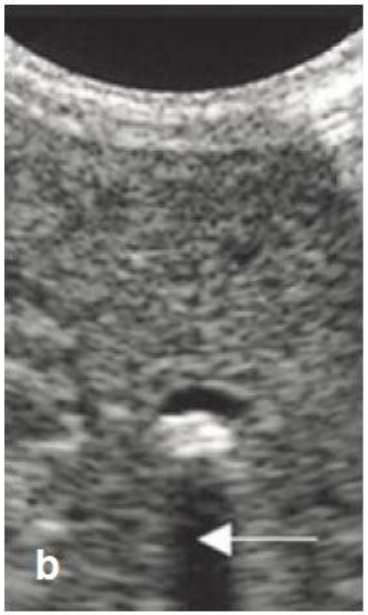




Image Acquisition

Image Acquisition

- In the past, all medical images were viewed and stored on **photographic** film.

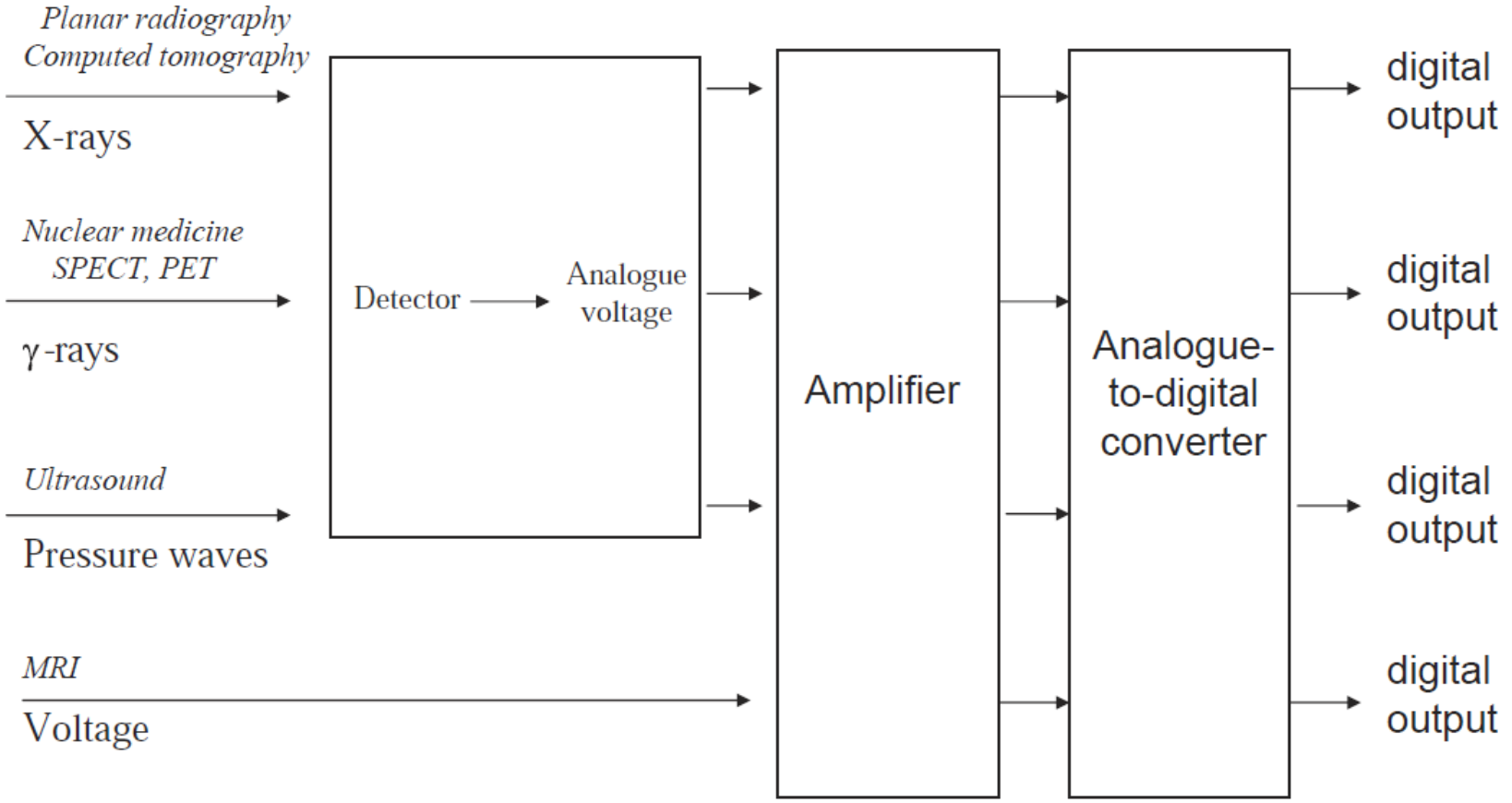
- At present, most images are acquired and stored digitally
  - They can be **post-processed** and **filtered** to maximize the diagnostic quality
  - Images acquired at different times can be quantitatively compared
  - Images can also be transferred **on-line** to other locations if required

Analogue-to-digital Converter (ADC)

- Takes an **analogue** input voltage from the medical imaging system and produces a **digital** output
  - Common requirement for producing digital images
  - Non-electrical signals must be converted into voltages using an appropriate detector
  - The ideal ADC converts a continuous voltage signal into a digital signal with as high fidelity and little noise as possible.
- Important specifications:
  - **Dynamic** range
  - **Voltage** range (maximum-to-minimum)
  - Maximum **sampling** frequency
  - Frequency **bandwidth**

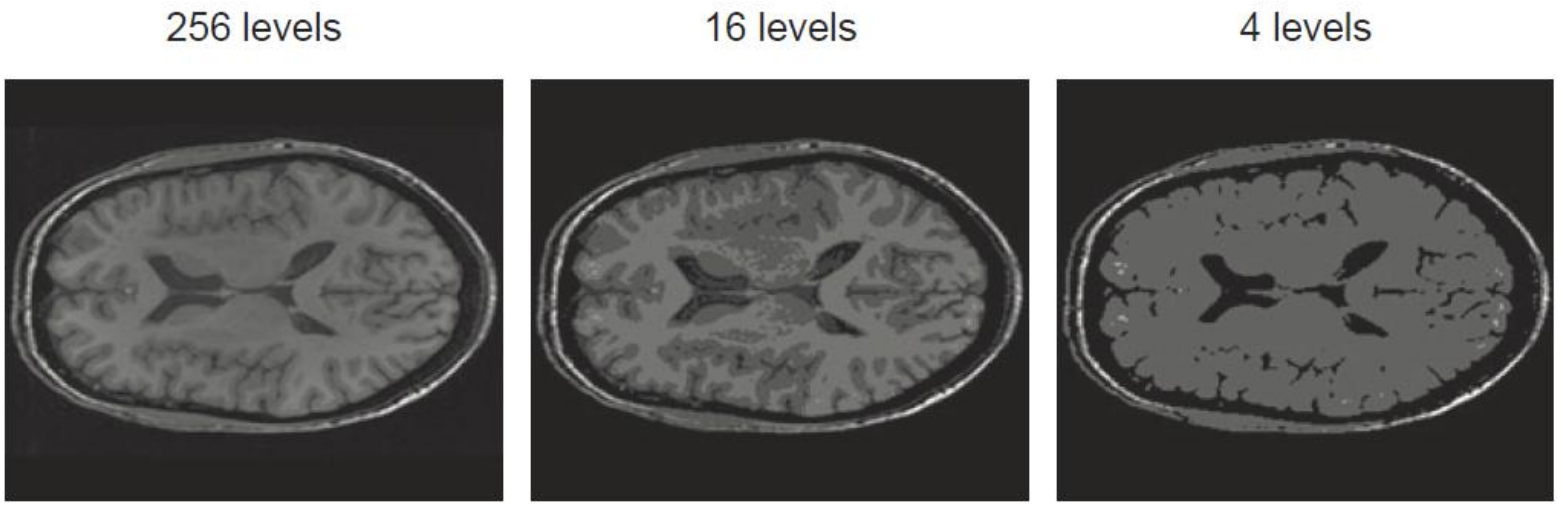
Analogue-to-digital Converter (ADC) cont.

- Data flow from different medical imaging modalities to produce a digital output.



Dynamic Range and Resolution

- The number of **colour** levels or the number of **gray** levels
  - Greater the number of levels, the higher the quality of the image
  - The gap between these levels is called the **resolution**
- Example: representation of an MRI image with a maximum of 256, 16 and 4 gray levels, corresponding to 8, 4 and 2-bit ADCs.

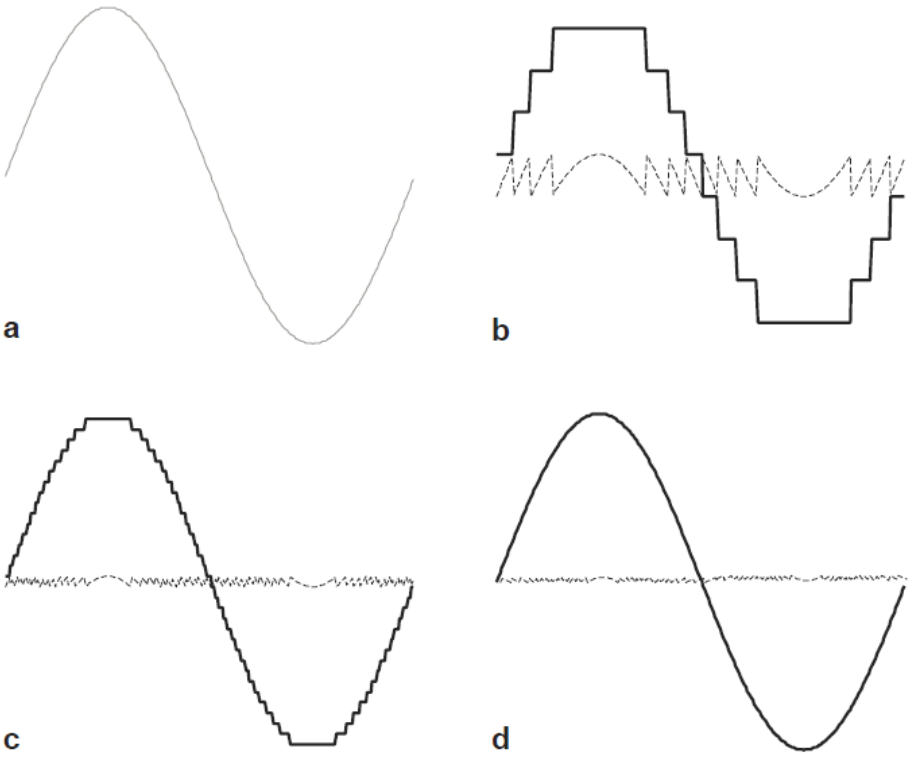


Quantization Error

- The difference between the true analogue input signal and the digitized output
  - Even ADCs with very high resolution cannot reproduce an analogue signal perfectly
  - Greater the number of bits, smaller this error becomes
- Always between 0 and $\pm 1/2$ of the ADC resolution
- To minimize the relative contribution of quantization errors
  - Input voltage to the ADC should be as large as possible, without 'saturating'
  - An amplifier is used with sufficient gain to give small signals to fill the ADC
  - Saturation will produce artifacts

Quantization Error cont.

- Example: Signal recorded by ADC (dark line) and quantization error (dashed line)
  - a: The analog sinusoidal signal
  - b: A three-bit ADC
  - c: A five-bit ADC
  - d: A six-bit ADC



Sampling Frequency and Bandwidth

- The Nyquist theorem states that a signal must be sampled at least **twice** as fast as the bandwidth of the signal to reconstruct the waveform accurately
  - Defines the **maximum frequency** that can be sampled and the **bandwidth** over which measurements are to be made
  - Otherwise, the high-frequency content will **alias** at a frequency inside the spectrum of interest

Sampling Frequency and Bandwidth cont.

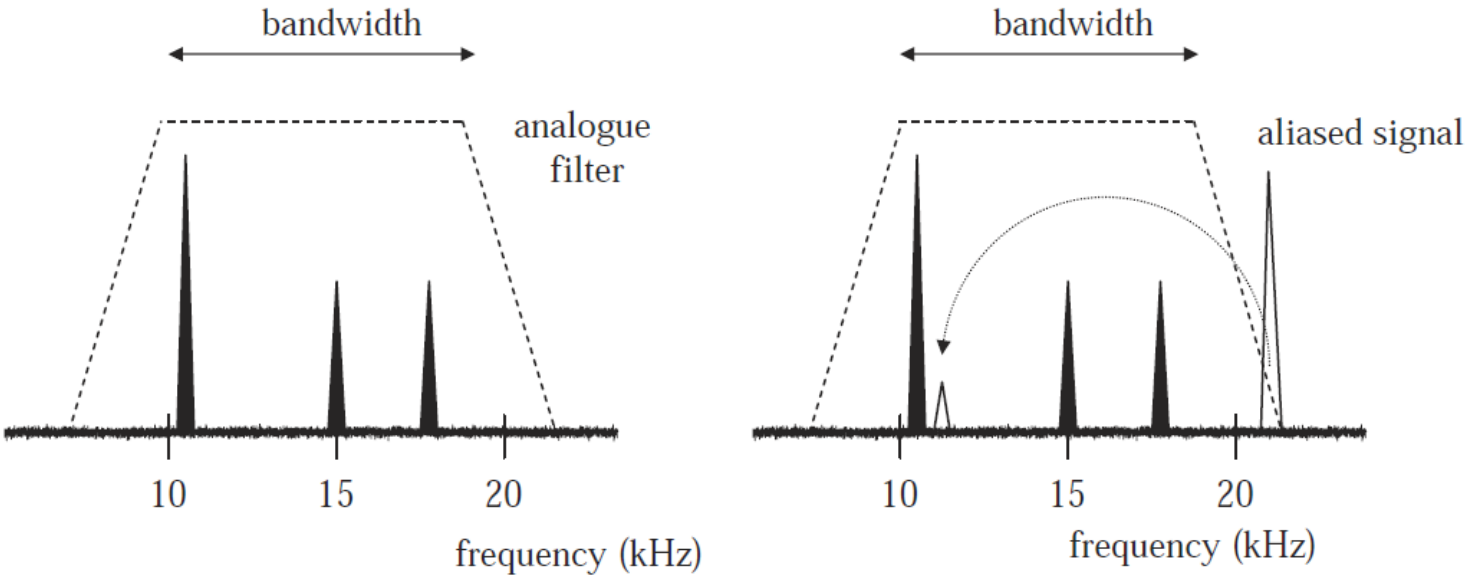
- Before digitization, data passed through either a **low-pass** or **band-pass** analogue filter to remove noise components outside the signal bandwidth
  - Acquiring a larger bandwidth than necessary is sub-optimal since the noise level is proportional to the square root of the bandwidth of the receiver
  - If the measurement bandwidth is less than the range of the frequencies that one is digitizing, the signal outside the measurement bandwidth will be **aliased** back into the spectrum
  - Each particular ADC has a specified maximum sampling rate at which it can achieve its highest resolution, Ex: 80 MHz with a 14 or 16-bit resolution
  - If the signal to be digitized has a frequency greater than the maximum sampling frequency of the ADC, then it must be **down-converted**

Sampling Frequency and Bandwidth cont.

- Example:

Left: A set of signals between 10 and 20 kHz is acquired by setting the central frequency of the ADC to 15 kHz with a 10 kHz bandwidth.

Right: An unexpected signal outside the bandwidth at 22 kHz aliased back into the spectrum. The intensity of the aliased signal is reduced compared to its true intensity since it is partially filtered out by the analogue filter.



Sampling Frequency and Bandwidth cont.

- Problems with the simple sampling scheme:
  - Analogue filters cannot be made absolutely '**sharp**' and have other problems such as **group delays** and frequency-dependent **phase shifts**
  - Resolution of a 14- or 16-bit ADC may not be sufficient to avoid **quantization** noise, but higher resolution ADCs typically do not have high enough sampling rates and are extremely expensive
  - **Oversampling** can solve these problems

Digital Oversampling

- Sampling the signal at a much higher frequency than required by the Nyquist theorem
  - Edges of the analogue filter are placed well away from the actual signals
  - Sharpness of the filters is no longer a problem
  - Other issues of group delays and phase shifts also disappear
- A digital filter is then applied to the sampled data to select only the bandwidth of interest
  - A digital filter can be designed to have an extremely **sharp** transition
  - Filtered data have N-times as many data points (where N is the oversampling factor)
- In the final step, successive data points are averaged together to '**decimate**' the data
  - Quantization noise in the decimated data set is **low** since the quantization error is **random**

Digital Oversampling cont.

- For every factor-of-four in oversampling, the equivalent resolution of the ADC increases by 1-bit
- Example: An illustration of digital oversampling

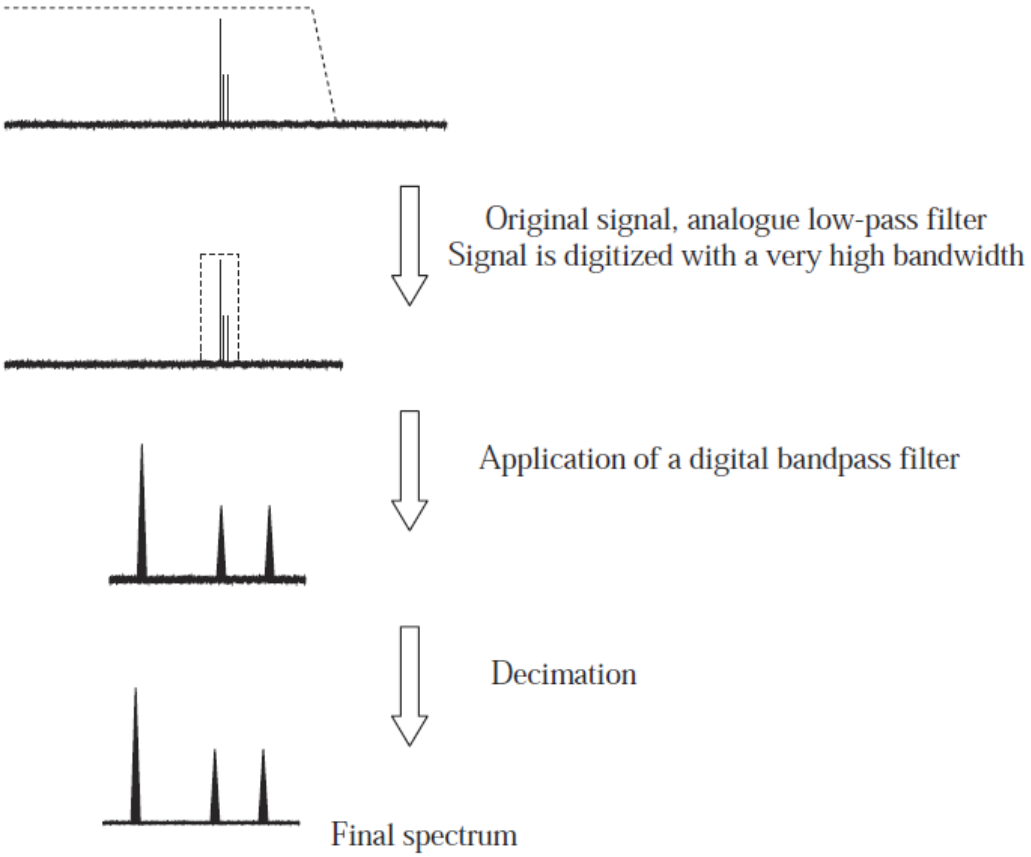


Image Filtering

- For further **enhancements** after the data have been acquired, digitized, and the images reconstructed
  - Mostly either **high-pass** or **low-pass**, referring to their characteristics in the **spatial frequency** domain
- High-pass filter
  - Accentuates the high spatial frequency components in the image
  - Small objects and edges in the images are '**sharpened**'
  - Improving the spatial resolution
  - Decreases the SNR of the filtered image
- Low-pass filter
  - Attenuates the noise and therefore, increases SNR of the filtered image
  - Smooths out other features represented by **high** spatial frequencies

Image Filtering cont.

- Example: effects of low-pass and high-pass filtering

